

AMENDMENT NO. 1 MARCH 2026

TO

IS 15351 : 2015 AGRO TEXTILES — LAMINATED HIGH DENSITY POLYETHYLENE (HDPE) WOVEN GEOMEMBRANE FOR WATER PROOF LINING — SPECIFICATION

(Second Revision)

(Page 2, Table 1) — Substitute the following for the existing:

Sl No.	Property	Requirement				Method of Test, Ref to
		Type I	Type II	Type III	Type IV	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Thickness at 24 KPa, mm, <i>Min</i>	0.25	0.50	0.75	1.00	IS 7016 (Part 1/Sec 3)
ii)	Mass (g/m ²), <i>Min</i>	250	420	650	950	IS 7016 (Part 1/Sec 2)
iii)	Dimensions (length and width)	As declared +1 percent with no negative tolerance				Annex B of IS 11652
v)	Breaking load on 20 cm × 10 cm strip, N, <i>Min</i> before U.V exposure:					IS 16635
	a) Warpway	3 300	6 600	8 200	10 700	
	b) Weftway	3 000	6 000	7 400	9 700	
vi)	Strain at maximum load, percent	20 ± 5				IS 16635
vii)	Breaking load on 20 cm × 10 cm strip, N, <i>Min</i> after U.V exposure of 1 000 h for warp and weft	80 percent of the actual original value				Annex B and IS 16635
ix)	Tear resistance, N, <i>Min</i>	100	250	300	350	Method A of IS 7016 (Part 3/Sec 1)
x)	Puncture resistance, N, <i>Min</i>	300	500	725	900	Annex D
xi)	Bursting strength, ball burst, N/cm ² , <i>Min</i>	60	110	135	200	IS 7016 (Part 6/Sec 1)

(Page 3, Annex A) — Delete the following references:

<i>IS No.</i>	<i>Title</i>
7016	Methods of test for coated and treated fabrics:
(Part 1) : 1982	Determination of roll characteristics (<i>first revision</i>)
(Part 3) : 1981	Determination of tear strength (<i>first revision</i>)
(Part 6) : 1984	Determination of bursting strength (<i>first revision</i>)
11652 : 2000	Textiles — Woven sacks for packing cement — High density polyethylene (HDPE)/polypropylene (PP) — Specification (<i>second revision</i>)

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<i>IS No.</i>	<i>Title</i>
13162 (Part 5) : 1992	Geotextiles — Methods of test: Part 5 Determination of tensile properties using a wide width strip'

(Page 3, Annex A) — Insert the following references at the appropriate place:

<i>'IS No.</i>	<i>Title</i>
IS 7016	Methods of test for coated and treated fabrics:
(Part 1)	Determination of roll characteristics,
(Sec 2) : 2019	Methods for determination of total mass per unit area, mass per unit area of coating and mass per unit area of substrate (<i>second revision</i>)
ISO 2286-2 : 2016	
(Sec 3) : 2019	Method for determination of thickness (<i>second revision</i>)
ISO 2286-3 : 2016	
IS 7016	Methods of test for rubber or plastics coated fabrics:
(Part 3/Sec 1) :	Determination of tear resistance, Section 1 Constant rate of tear methods (<i>third revision</i>)
2022/ISO 4674-1 :	
2016	
(Part 6/Sec 1) :	Determination of bursting strength, Section 1 Steel-ball method (<i>third revision</i>)
2022/ISO 3303-1 :	
2020	
IS 11652 : 2017	Textiles — High density polyethylene (HDPE)/polypropylene (PP) woven sacks for packaging of 50 kg cement — Specification (<i>third revision</i>)
IS 16635 : 2017	Geosynthetics — Wide-width tensile test'
ISO 10319 : 2015	

(Page 4, clause **B-1**, line 7) — Substitute 'IS 16635' for 'IS 13162 (Part 5)'.

(Page 4, clause **B-5.3**, line 4) — Substitute 'IS 16635' for 'IS 13162 (Part 5)'.

(Page 4, clause **B-5.4**, line 4) — Substitute 'IS 16635' for 'IS 13162 (Part 5)'.

(Page 6, Annex D) — Substitute the following for the existing:

'ANNEX D

[Table 1, *SI No.* (x)]

TEST METHOD FOR INDEX PUNCTURE RESISTANCE

D-1 PRINCIPLE

A test specimen is clamped without tension between circular plates of a ring clamp attachment secured in a tensile testing machine. A force is exerted against the centre of the unsupported portion of the test specimen by a solid steel rod attached to the load indicator until rupture of the specimen occurs. The maximum force recorded is the value of puncture resistance of the specimen.

D-2 APPARATUS

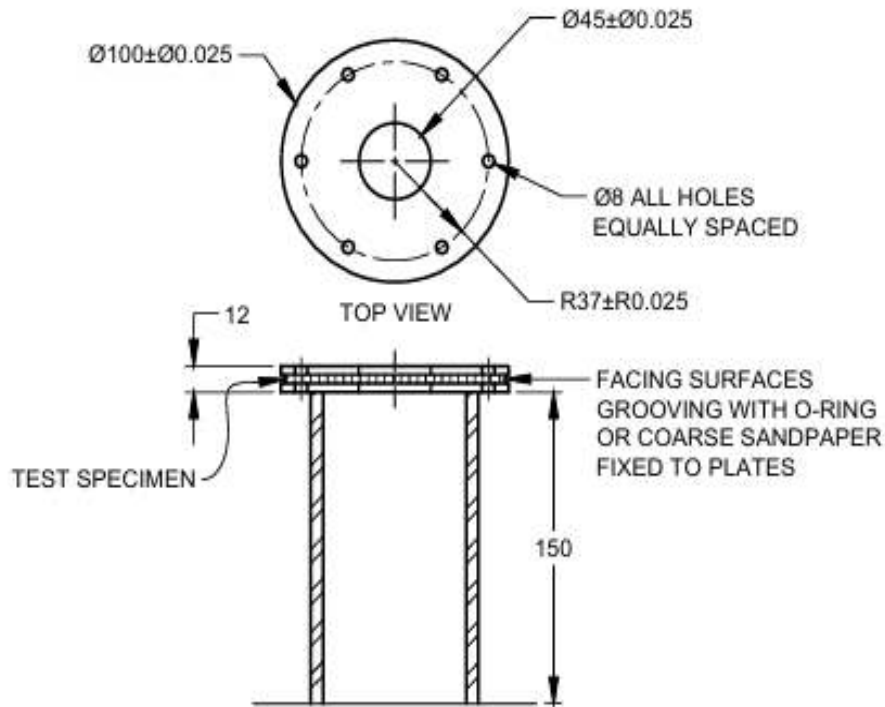
D-2.1 Tensile/Compression Testing Machine — of the constant-rate-of extension (CRE) type.

D-2.2 Ring Clamp Attachment — consisting of concentric plates with an open internal diameter of 45 mm $\pm$ 0.025 mm capable of clamping the test specimen without slippage. A suggested clamping arrangement is shown in Fig. 1. The external diameter is suggested to be 100 mm $\pm$ 0.025 mm. The diameter of the six holes used for securing the ring clamp assembly is suggested to be 8 mm and equally spaced at a radius of 37 mm. The

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surfaces of these plates can consist of grooves with a-rings or coarse sandpaper bonded onto opposing surfaces.

D-2.3 Solid Steel Rod — Solid steel rod, with a diameter of $7.92 \text{ mm} \pm 0.25 \text{ mm}$ and a thickness of $0.8 \text{ mm} \pm 0.1 \text{ mm}$ at the extreme tip having a flat chisel tip with both side bevel (*see* Fig. 2). The tip edges shall be rounded to a nominal 0.25 mm radius. The piercing instrument shall be oriented to intercept the warp and weft threads of the mounted specimen at an angle of approximately 45° .



All dimensions in millimetres.

FIG. 1 TEST FIXTURE DETAIL (NOT TO SCALE)



FIG. 2 SHAPE OF TEST PROBE

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D-3 SAMPLING

D-3.1 Laboratory Sample

For preparation of the laboratory sample, a swatch extending across the full width of the product and of sufficient length along the selvedge shall be taken from each sample roll so as to meet the requirements specified in **D-3.2**.

D-3.2 Test Specimens

Select from the laboratory sample, sufficient number of samples each having a minimum diameter of 100 mm to facilitate clamping. Space the specimens along a diagonal on the unit of the laboratory sample. Take no specimens nearer the selva ge or edge of the bed.

D-4 CONDITIONING

Bring the specimens to moisture equilibrium in the atmosphere for testing puncture resistance (65 percent $\pm$ 5 percent relative humidity and 27 °C $\pm$ 2 °C temperature). Equilibrium is considered to have been reached when the increase in the mass of the specimen, in successive weightings made at intervals of not less than 2 h, does not exceed 0.1 percent of the mass of the specimen.

D-5 PROCEDURE

D-5.1 Select the load range of the tensile/compression testing machine such that the rupture occurs between 10 percent and 90 percent of the full-scale load.

D-5.2 Centre and secure the specimen between the holding plates ensuring that the test specimen extends to or beyond the outer edges of the clamping plates.

D-5.3 Test at a machine speed of 300 mm/min $\pm$ 10 mm/min until the puncture rod completely ruptures the test specimen.

NOTE — The rate of testing specified is not an indication of the performance of the specimen for its end use.

D-6 CALCULATION

Calculate the average puncture resistance and standard deviation for all tests as read directly from the recording instrument.'

(Page 7, clause F-1, line 4) — Substitute '590 °C' for '550°C'.

(Page 7, clause F-1, line 5) — Substitute '590 °C' for '550°C'

(Page 7, clause F-2.2, line 2) — Substitute '3 g' for '30 g'.

(Page 8, clause F-2.5, line 2) — Substitute '590 °C' for '550 °C'.

(Page 8, clause F-4.1, line 1) — Substitute '590 °C' for '550 °C'.

(Page 8, clause F-4.3, line 1) — Substitute '3 g $\pm$ 0.2 g' for '30 g $\pm$ 2 g'.

(Page 8, clause F-4.5, line 2) — Substitute '590 °C' for '550 °C'.